GENAI · RETRIEVAL-AUGMENTED GENERATION

# Advanced RAG Portfolio Project

A production-style Retrieval-Augmented Generation application you can build, debug, evaluate — and confidently explain in an interview.

Python | LangChain | OpenAI | FAISS | Pinecone | BM25 | Cross-Encoder | Streamlit

01

# What You Build

- Upload and process **PDF, TXT and DOCX documents**
- Build and **update a vector database** without re-embedding everything
- Choose between **local FAISS** and **remote Pinecone**
- Ask questions through a **Streamlit chat interface**
- Get answers with **source citations** (file, page, section)
- Maintain **conversation memory** for follow-up questions
- Inspect the full pipeline with **Retrieval Debug Mode**
- Measure quality with a **basic evaluation framework**

02

# Advanced RAG Techniques You Learn

### Parent-Child Chunking

- Small child chunks for precise retrieval
- Large parent chunks for richer context

### Multi-Query Retrieval

- LLM writes several query variations
- Better recall and follow-up handling

### Hybrid Retrieval

- Vector search plus BM25 keywords
- Fused with Reciprocal Rank Fusion

### Two-Stage Cross-Encoder Reranking

- Fast retrieval, then accurate reranking
- Bi-encoder vs cross-encoder trade-off

### Context Refinement

- Drop irrelevant retrieved chunks
- Apply a context and token budget

THE COMPLETE PIPELINE

Ingestion > Chunking > Embedding > Indexing > Retrieval > Reranking > Refinement > LLM > Citations

03

# Engineering Skills & Tech Stack

## TECHNOLOGIES

Python | LangChain | OpenAI Chat + Embeddings | FAISS | Pinecone | BM25 | Cross-Encoder / Sentence Transformers | Streamlit

## ENGINEERING SKILLS

- Document ingestion and metadata handling
- Vector database management
- Build / update / clear index workflows
- LLM-powered query transformation
- Retrieval pipeline design
- Conversation memory with LangChain
- Source citation and grounding
- Context and token-budget management
- Basic RAG evaluation
- Offline unit testing
- Environment and configuration management
- Debugging a retrieval pipeline end to end

**Build, debug, evaluate and improve a real-world RAG system — and explain why each technique exists.**

Not just another chatbot: the entire retrieval pipeline behind a production-style GenAI application.